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Advanced COVID-19 detection using cough signals with space reconstruction and 3D deep convolutional neural networks

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This study proposes a novel and robust methodology for detecting COVID-19 through the analysis of cough audio signals by integrating advanced feature engineering with deep learning techniques. We utilize Phase Space Reconstruction (PSR) to convert raw cough sound signals into a multidimensional feature space that captures the complex dynamic behavior inherent in the data. These features are further encoded into a three-dimensional tensor representation, which serves as input to a custom-designed deep convolutional neural network (3D DCNN) comprising five convolutional layers with max-pooling operations. Our model performs multi-class classification to distinguish between COVID-19 positive cases, symptomatic non-COVID individuals, and healthy controls. Leveraging the extensive COUGHVID dataset, which contains over 8,400 cough recordings from diverse populations, the proposed approach achieves a classification accuracy of 98.5%, a recall of 96.5%, and a specificity of 99.7%, outperforming current state-of-the-art methods. The results demonstrate that synthesizing dynamic signal representations with hierarchical 3D deep learning architectures significantly enhances diagnostic accuracy. This non-invasive and scalable framework offers promising potential for rapid and reliable COVID-19 screening, contributing valuable support to public health surveillance and clinical decision-making.

Keywords COVID-19 detection, Cough sound analysis, Acoustic signal processing, PSR, Feature engineering, 3D tensor representation

Coronavirus disease 2019 (COVID-19), caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), was first identified in late 2019 and subsequently declared a global pandemic by the World Health Organization (WHO) on February 11, 2020. This novel virus shares genetic and clinical characteristics with other coronaviruses responsible for prior outbreaks, namely the severe acute respiratory syndrome (SARS) epidemic in 2002 and the Middle East respiratory syndrome (MERS) outbreak in 2012. According to WHO reports as of August 18, 2024, there have been over 776 million confirmed cases globally, with a death toll exceeding 7 million. Clinical observations indicate that approximately 5% to 20% of infected individuals develop severe symptoms requiring intensive care, with mortality rates ranging between 26% and 62%, influenced by factors such as comorbidities and healthcare accessibility[1–5].

Conventional diagnostic approaches have increasingly incorporated artificial intelligence (AI) techniques to detect COVID-19 from medical imaging data, including chest X-rays and CT scans, demonstrating promising performance. However, these imaging modalities require specialized equipment, trained personnel, and in some instances, expose patients to ionizing radiation. In contrast, the analysis of cough and breath sounds emerges as a non-invasive, cost-effective, and accessible alternative diagnostic approach. Recent efforts leveraging machine learning and signal processing methods have sought to identify distinctive acoustic biomarkers associated with COVID-19 infection from respiratory audio recordings. These studies underscore the feasibility of automated, scalable screening tools based on audio analysis that can complement existing diagnostic frameworks[6–10].

AI-driven analysis of chest X-rays and CT scans has indeed shown great promise for COVID-19 detection, with works like COVID-Net[11] and DenseCapsNet[12] demonstrating high accuracy. However, as noted in reviews such as[13], these methods rely on expensive, non-portable equipment and expose patients to ionizing radiation (in the case of CT). Our audio-based approach, in contrast, offers a truly non-invasive, zero-radiation, and

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highly accessible alternative that can be deployed using ubiquitous consumer devices like smartphones, making it more suitable for rapid, large-scale screening.

Overview of existing approaches

A critical factor in the development of cough-based COVID-19 detection systems is the availability of public datasets. These datasets, typically collected via crowd-sourcing platforms or clinical settings, vary significantly in size, demographic representation, and the types of audio recordings collected. Table 1 provides a summary of the key datasets commonly used in the literature, highlighting their scope and scale. For this study, we utilized the COUGHVID dataset due to its large size, public accessibility, and inclusion of multiple classes (COVID-19 positive, symptomatic negative, and healthy), which aligns with our multi-class classification objective.

Classic machine learning methods

In traditional machine learning approaches, a primary challenge lies in identifying discriminative features capable of effectively distinguishing between healthy and pathological respiratory sounds. For instance, investigates alterations in cough patterns among COVID-19 infected individuals utilizing a Hidden Markov Model (HMM)-based speech recognition framework, supplemented by formant frequency and pitch analyses. The proposed HMM-based cough recognition system consists of five HMM states, eight Gaussian Mixture Model (GMM) components, and employs Mel-Frequency Cepstral Coefficients (MFCCs) forming a 39-dimensional feature vector. Comparative analysis between COVID-19 positive and healthy cohorts revealed a 6.7% differential in recognition rates. Notably, significant variations were detected in formant frequencies F1, F3, and F4, whereas fundamental frequency (F0) and F2 exhibited less pronounced changes. These results validate the potential of the HMM-based system in accurately discriminating COVID-19-related coughs[19].

A subset of research efforts has also focused on improving system performance through optimized feature selection and classifier design. For example, proposes an ensemble-based multi-criteria decision-making framework to identify the most effective machine learning models for COVID-19 cough classification. By integrating four publicly available datasets—Cambridge, Coswara, Virufy, and NoCoCoDa—and employing ensemble techniques combined with TOPSIS ranking and entropy-based weighting, the study refines model selection processes. Notably, the Extra-Trees classifier achieved exemplary performance metrics, with an area under the ROC curve (AUC) of 0.95, precision of 1.0, and recall of 0.97, outperforming prior state-of-the-art approaches[20].

Furthermore, investigates the application of symbolic learning methods, specifically interval temporal logic-based decision trees and forests, for classifying COVID-19 from cough and breath recordings. These audio signals, treated as multivariate time series, were obtained from the University of Cambridge dataset. The study demonstrates that symbolic learning algorithms not only outperform leading non-symbolic classifiers on this dataset but also exceed the performance of most established methods applied to alternative datasets. Importantly, symbolic models provide explicit, interpretable rules, offering clinicians transparent insights into the characteristic features of COVID-19 positive respiratory sounds, thereby facilitating improved clinical decision support[21].

Beyond time-frequency and texture features, another line of research explores higher-order spectral analysis to capture non-Gaussian and nonlinear characteristics. For instance, a study by[22] utilized bispectral and entropy features extracted from cough signals, achieving high accuracy. The bispectrum, a third-order statistic, is particularly powerful for characterizing nonlinear phase couplings between frequency components that are not visible in the power spectrum. While both bispectrum and our PSR aim to capture nonlinear dynamics, they operate on fundamentally different principles: PSR reconstructs the state space of the underlying dynamical system in the time domain, revealing its geometric properties, whereas the bispectrum operates in the frequency domain. A comparative analysis of these complementary approaches could be a fruitful direction for future work. We conducted our experiments using the COUGHVID dataset[14], a large-scale, crowd-sourced repository. As summarized in Table 2, COUGHVID is among the largest publicly available datasets, which provides a robust foundation for model development.

Dataset name	Primary modalities	Total Samples	COVID-19 Positive Samples	Key characteristics & notes	Primary references
COUGHVID	Cough, Breathing, Speech	> 20,000	~ 2,000	Large-scale, crowd-sourced; includes self-reported symptoms and metadata (age, gender, location). Contains label noise inherent to crowd-sourcing.	14
Coswara	Cough, Breathing, Speech, Vowel sounds	~ 2,500	~ 1,200	Crowd-sourced; includes detailed metadata and multiple respiratory sound types. Data collection was phased throughout the pandemic.	15
Cambridge	Cough, Breathing	~ 600	~ 300	Smaller, crowd-sourced dataset; used in several early studies on COVID-19 audio analysis.	16
Virufy	Cough	~ 1,200	~ 600	Crowd-sourced; focus on creating a globally diverse dataset for COVID-19 screening.	17
NoCoCoDa	Cough	~ 1,400	~ 400	The Noisy COVID-19 Cough Dataset; designed to include background noise to test model robustness.	18
MIT Open Voice	Cough, Vowel sounds, Sentence speech	~ 5,000	~ 200	Large dataset but with a relatively smaller number of confirmed COVID-19 cases.	MIT Lincoln Laboratory

Table 1. Summary of key public datasets for COVID-19 cough audio analysis.

Method	Feature/Approach	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)
[22]	Bispectral & Entropy Features	92.9	93.0	92.8	–	–
Proposed Method	PSR + 3D DCNN	98.5	96.8	96.5	99.7	96.7

Table 2. Comparative classification performance of the proposed method with state-of-the-art techniques. Note: The results for[22] are as reported in their publication. A dash (–) indicates the metric was not reported.

Recent studies have also explored higher-order spectral analysis, such as the complex bispectrum, for COVID-19 cough detection[23,24]. These methods are powerful for characterizing non-Gaussian and nonlinear phase couplings between frequency components. While both bispectrum and PSR aim to capture nonlinear dynamics, they operate on different principles. PSR reconstructs the state space of the underlying dynamical system directly from the time series, capturing geometric and topological properties of the signal's attractor. In contrast, the bispectrum is a frequency-domain method. A potential limitation of bispectral analysis is its computational cost and the challenge of interpreting the resulting high-dimensional features. Our PSR approach, by mapping the signal into a geometric space, provides a representation that is inherently suitable for spatial feature learning with CNNs, which may offer a more intuitive and compact characterization of the cough's dynamic pathology.

Deep learning based methods

Deep learning models are highly effective in both feature extraction and classification tasks due to their unified end-to-end learning paradigm. Leveraging a large number of trainable parameters and sophisticated mathematical operations, these architectures automatically learn hierarchical feature representations across multiple network layers. Initial layers typically capture low-level acoustic features, while deeper layers progressively identify more abstract, high-level characteristics, culminating in a final classification output. This end-to-end approach allows the neural network to directly process raw input signals and produce corresponding class labels without requiring manual feature engineering. Consequently, a critical research focus has been identifying which deep learning architectures best address the unique challenges associated with COVID-19 detection from cough audio.

Proposed a deep learning framework utilizing the Mini VGGNet architecture combined with segmentation techniques for improved COVID-19 detection. Evaluation on datasets from the University of Cambridge, Coswara, and NIH Malaysia revealed that segmentation significantly enhances model performance, whereas data augmentation had negligible effect. The model attained a test accuracy of 92.1%, AUC of 97.3%, precision of 91.0%, and recall of 91.0%, underscoring the importance of segmentation and dataset diversity in AI-driven COVID-19 diagnostics.

Addressing delays in conventional methods, developed a novel ensemble deep learning model incorporating Empirical Mean Curve Decomposition (EMCD) for signal decomposition, followed by extraction of MFCC, spectral, and statistical features. Feature fusion and optimization were performed using the Modified Cat and Mouse Based Optimizer (MCMBO), feeding into an optimized deep ensemble classifier (ODEC) that combines Radial Basis Function (RBF), LSTM, and deep neural networks (DNN). Validation demonstrated 96% accuracy and 92% precision, emphasizing the method's efficacy for early COVID-19 detection.

Beyond convolutional and recurrent architectures, Vision Transformers (ViTs) and adaptive mechanisms have shown remarkable success in medical image analysis. For instance, methods like regularized transformers with adaptive token fusion and large adaptive filters with report-guided alignment have been developed for Alzheimer's disease and tuberculosis/pneumonia diagnosis from brain MRIs and chest X-rays, respectively. These approaches excel at capturing long-range dependencies and integrating multimodal clinical information. Similarly, architectures like CTBViT demonstrate the efficacy of efficient ViT blocks for tuberculosis classification. While these methods are tailored for imaging data, their success underscores the potential of self-attention mechanisms. Our work, in contrast, focuses on capturing the nonlinear dynamics of audio signals through PSR, providing a rich, physics-inspired feature space for a 3D-CNN to exploit. Future work could explore merging the long-range contextual modeling of transformers with the dynamic feature extraction of PSR for audio-based diagnosis[25–30].

Similarly, proposed segmenting cough sounds into overlapping frames identified using YAMNet, followed by fractal dimension analysis to generate image representations classified by a pretrained Vision Transformer (ViT) model. Tested on COUGHVID, VIRUFY, and COSWARA datasets, the method reached accuracies of 98.45%, 98.15%, and 97.59%, respectively, surpassing prior works and offering a high-performance, contactless COVID-19 screening tool.

Material and method

The proposed approach for COVID-19 detection from cough audio consists of two principal components: feature engineering and a deep convolutional neural network (CNN) architecture. The initial phase focuses on transforming the raw cough audio signals into a feature-rich and more interpretable representation to enhance the efficacy of subsequent classification. This is achieved through PSR, a nonlinear dynamic systems technique that maps one-dimensional time-series data into a multidimensional phase space, effectively revealing the underlying 'fingerprint' of the system's dynamics. This transformation effectively enriches the input data, producing a multidimensional format that is highly suitable for learning by deep neural networks. Following feature engineering, the reconstructed data is fed into a specially designed deep CNN model that automatically extracts hierarchical features and performs classification. The network architecture is optimized to leverage the complex spatial and temporal information embedded within the multidimensional PSR representation. Through sequential convolutional layers, nonlinear activations, and pooling operations, the CNN learns discriminative patterns associated with COVID-19-related cough characteristics. This enables robust differentiation between COVID-19 positive, symptomatic non-COVID, and healthy cough samples. The 3D convolutional architecture was chosen specifically to leverage the spatial correlations within the volumetric PSR tensor. Unlike 2D CNNs designed for images or RNNs for sequences, the 3D DCNN can learn features that capture the complex three-dimensional geometry of the phase space attractor, which is central to our dynamic feature representation.

Figure 1. Schematic of the proposed COVID-19 detection framework. The process begins with raw cough audio acquisition, which is transformed into a multidimensional feature space using PSR. The resulting dynamics are encoded into a 3D tensor, which serves as the input to a custom 3D Deep Convolutional Neural Network (DCNN) for final classification into COVID-19 positive, symptomatic non-COVID, or healthy categories.

The COUGHVID dataset is a large-scale, crowd sourced repository designed to facilitate cough analysis research, particularly for COVID-19 diagnosis. It contains over 20,000 cough recordings paired with metadata such as age, gender, geographic location, and COVID-19 diagnosis status.

Following the approach recommended by previous studies, we applied a quality control filter to exclude samples with a cough probability score below 0.5, ensuring that only reliable cough recordings are used for model training and evaluation. This preprocessing step narrowed the dataset to 8,407 high-quality samples, consisting of 6,466 healthy individuals, 658 COVID-19 positive cases, and 1,283 symptomatic but COVID-negative cases.

This curated dataset provides a balanced and robust foundation for training and validating cough-based COVID-19 detection algorithms, enhancing the reliability of subsequent classification results. To mitigate issues of noisy data and non-cough recordings inherent in a crowd-sourced dataset, we applied a quality control filter to exclude samples with a cough probability score below 0.5. While this improves data reliability, it does not eliminate the possibility of mislabeled COVID-19 status, a common limitation of crowd-sourced data.

In addition PSR is a powerful nonlinear dynamics technique employed to transform raw one-dimensional time-series signals into a higher-dimensional space that reveals their underlying temporal structures and dynamic behaviors. By mapping the original cough audio signal into a multidimensional phase space, PSR captures intrinsic signal characteristics that are often hidden in conventional time-domain or frequency-domain analyses. This method constructs vectors from time-delayed copies of the original signal, defined by key parameters: the embedding dimension and time delay. Selecting appropriate values for these parameters is essential to unfold the signal's state space effectively, thereby exposing intricate dynamical patterns caused by COVID-19 related respiratory changes. Such enriched representations provide a more informative and discriminative feature set, enhancing the ability of machine learning models to differentiate between COVID-19 positive, symptomatic, and healthy cough sounds.

The multidimensional output generated by PSR can be further processed into structured formats compatible with advanced deep learning models. In this study, the phase space trajectories derived from cough audio are

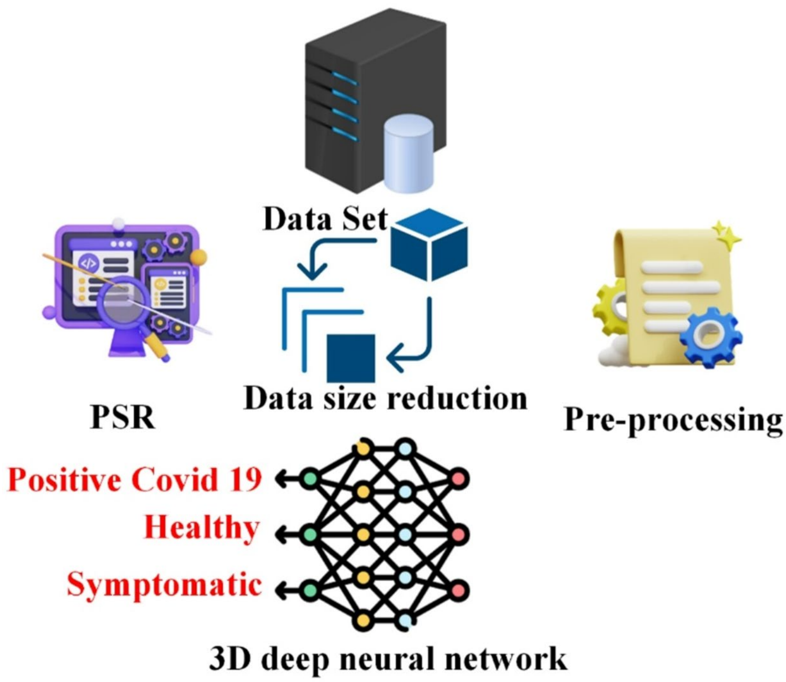

Fig. 1. provides a comprehensive flow diagram illustrating the end-to-end process of the proposed COVID-19 detection system, from raw audio acquisition through feature transformation via PSR to final classification by the deep CNN model.

encoded into three-dimensional tensors, providing a spatial-temporal volume of data that preserves complex geometric and temporal relationships within the signal. This transformation enables the subsequent deep convolutional neural network to exploit hierarchical feature extraction more effectively, capturing both local variations and global patterns unique to COVID-19 infection. As demonstrated by the distinct scattering distributions seen in phase space representations of infected versus healthy coughs, PSR facilitates a robust encoding of subtle acoustic biomarkers, thereby significantly improving diagnostic accuracy and reliability.

PSR is a valuable technique for examining a system's dynamics within a multidimensional framework. The RPS is derived from the system's output signal, S[n], $n=1, 2, 3,\ldots, N$. By defining the vector as Eq. (1). Additionally, the RPS is formed by concatenating such vectors for all possible values of n (Eq. (2)). Where T denotes the matrix transpose, each row of the RPS corresponds to a possible system condition. The time lag, τ, and the dimensionality of the space, d, are two crucial parameters to be determined in RPS. τ can be determined using mutual information in Eq. (3).

The time lag, τ, was determined using the mutual information (MI) criterion, selecting the first minimum of the average MI across all samples in the COUGHVID dataset to ensure maximum independence between delayed coordinates. This voting procedure yielded $\tau=13$. The embedding dimension, d, was selected empirically by evaluating model performance for d ranging from 3 to 8, as detailed in Sect. 4. The optimal performance was achieved at $d=5$, which is consistent with the theoretical condition $d\geq 2FD$ (where FD is the fractal dimension) for successfully unfolding the dynamics of a chaotic system.

$$\overline{S_n}=\left[S_n,S_{n+\tau},S_{n+2\tau},\ \ldots,S_{n+(d-1)\tau}\right] \tag{1}$$

$$RPS=\left[\begin{array}{cccc}\overline{S_1} & \overline{S_2} & \ldots & \overline{S_n}\end{array}\right]^T \tag{2}$$

$$MI\left(X,Y\right)=\sum_{y\in Y}\sum_{x\in X}P_{(X,Y)}\left(x,y\right)\log\left(\frac{P_{(X,Y)}\left(x,y\right)}{P_{(X)}\left(x\right)P_{(Y)}\left(y\right)}\right) \tag{3}$$

By the joint probability distribution of X and Y, p(x, y), and the marginal probability distributions of X and Y, p(x) and p(y), MI(X, Y) determines the mutual information between two variables X and Y. Now, assume X and Y respectively as x[n] and x[n + τ], choosing the first minimum of MI guarantees that x[n] and x[n + τ], as the two dimensions of the space, have the maximum independence, while lower and higher τ values lead to high dependence and irrelevance, respectively[14]. By choosing this optimized τ value, the data is more likely to scattered as much as possible in RPS. Here, we calculate the MI of all samples within the COUGHVID database. Using a voting procedure, τ = 13 has been chosen in our experiments.

false nearest neighbor algorithm to determine a proper value for d[14]. However, d ≥ 2FD is a sufficient, but not necessary condition for choosing an appropriate value for d, wherein FD is the fractal dimension of the signal. Here, we have selected the optimum d value by experiments as described in Sect. 4.

The topological characteristics of a system's RPS have been demonstrated to hold valuable information about the system. The RPS method is commonly used in time-series analysis to extract spatial features from data, which can be further processed by conventional machine learning algorithms and classifiers. In contrast, some researchers propose extracting 2D chaogram images from the RPS and using them with DNNs for automatic feature extraction and classification. As described in Deep Learning with Python by[14], feature engineering involves manipulating input data to facilitate the classifier model's training process. In this context, the PSR can be viewed as a feature engineering tool, aimed at representing raw audio data in a more interpretable, feature-rich format. Figure 2 shows two COVID-19, and healthy cough signals and their RPS representations.

As illustrated in Fig. 2, the PSR representation of cough signals from COVID-19 positive cases exhibits a dispersed scattering pattern characterized by a high density of irregular points, which is notably absent in healthy individuals. This distinctive pattern underscores the effectiveness of PSR in capturing the complex dynamical behavior inherent to pathological cough sounds. By transforming the raw audio into this enriched multidimensional space, PSR facilitates the extraction of discriminative features critical for differentiating COVID-19 related coughs from healthy ones.

In the present study, a three-dimensional deep convolutional neural network (3D DCNN) is specifically designed to leverage this PSR-derived representation, enabling automatic learning of relevant spatial-temporal features for robust classification. To align with the network's input requirements, the PSR output is systematically quantized and formatted into a three-dimensional tensor structure, as detailed in the following, thereby ensuring seamless integration with the deep learning framework.

The PSR is a continuous, d-dimensional representation of the underlying system dynamics. To facilitate effective analysis and reduce computational complexity, Principal Component Analysis (PCA) is first applied to the RPS data, reducing its dimensionality to three while preserving the majority of the signal's variance. This dimensionality reduction highlights the most significant features, such as attractors and trajectories, rendering the complex dynamics more interpretable in a 3D coordinate system. Subsequently, this 3D phase space is discretized into a uniform grid of $256\times256\times256$ cells. Each point in the phase space is quantized using 8-bit resolution along each axis, assigning values between 0 and 255. The frequency of points falling into each cell is computed, thereby generating a 3D tensor that encodes the spatial distribution of the reconstructed dynamics. This tensor serves as the input to the 3D deep convolutional neural network (DCNN). While increasing the grid resolution beyond 256 cells per axis can mitigate quantization errors and capture finer structural details, it imposes substantially higher computational costs. Through empirical evaluation, a resolution of 256 was found to provide an optimal trade-off between preserving critical spatial features and maintaining feasible computational efficiency. So conversion process begins by representing the audio signal in the RPS format. The

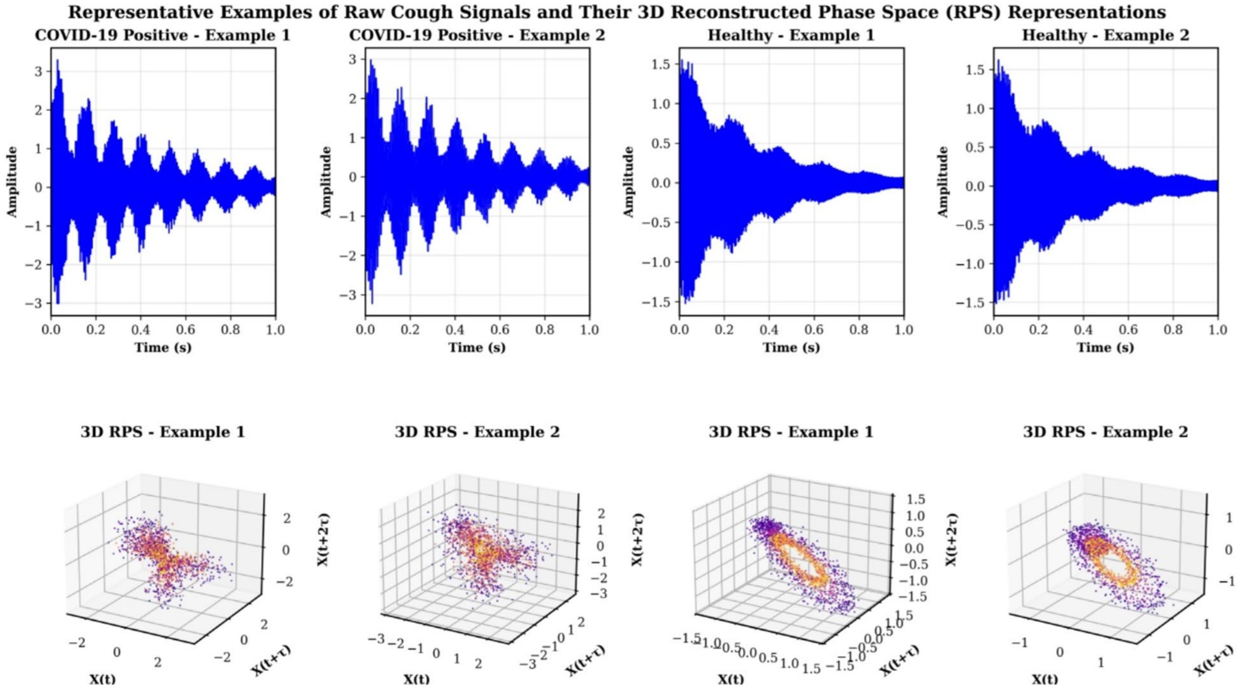

Fig. 2. Representative examples of raw cough audio signals (top row) and their corresponding 3D PSR representations (bottom row) for two COVID-19 positive (left column) and two healthy (right column) samples from the COUGHVID dataset. The dispersed, complex attractor geometry in the COVID-19 RPS plots contrasts with the more regular patterns in healthy coughs, highlighting the discriminative power of PSR.

RPS is divided into 256 grids along each axis, and the frequency of points within each grid cell is calculated to construct the 3D tensor.

In this study, we designed and implemented a three-dimensional deep convolutional neural network (3D DCNN) tailored to classify volumetric tensor representations derived from cough audio signals into normal and abnormal categories. The detailed architecture of the proposed 3D DCNN model, including its convolutional layers, activation functions, and pooling operations, is depicted in Fig. 3.

Principal Component Analysis (PCA) was chosen over supervised methods like Linear Discriminant Analysis (LDA) for its unsupervised nature, which preserves the intrinsic geometric structure of the phase space without biasing the features based on training labels, potentially leading to better generalization. An experimental comparison with LDA resulted in a slightly lower accuracy of 97.8%, supporting the choice of PCA.

The proposed 3D Deep Convolutional Neural Network (DCNN) architecture comprises five sequential convolutional layers, each employing a $3 \times 3 \times 3$ kernel size. These convolutional layers are followed by max-pooling operations, which downsample the spatial dimensions of the feature maps using a $2 \times 2 \times 2$ pooling window with a stride of 2. The number of filters increases progressively across the convolutional layers, starting from 64 in the first layer and doubling thereafter to 128, 256, and 512 filters, with the final two convolutional layers maintaining 512 filters. This gradual increase in filter depth enables the network to learn from low-level to complex high-level spatial features effectively.

After every convolutional layer, a Rectified Linear Unit (ReLU) activation function is applied to introduce non-linearity, which aids in learning complex feature representations and mitigates issues related to vanishing gradients. The decision to use a relatively small $3 \times 3 \times 3$ kernel size is motivated by its ability to capture fine-grained spatial details within a localized receptive field. This kernel dimension helps preserve spatial resolution, reduces the total number of trainable parameters, and thus lowers the risk of overfitting. Moreover, smaller kernels foster computational efficiency, facilitating faster convergence and enabling the stacking of multiple convolutional layers to hierarchically extract intricate feature patterns from the 3D tensor input.

The input to the network is a 3D tensor of size $256 \times 256 \times 256$, directly derived from the PSR representation without any resizing or interpolation, thereby preserving the full spatial integrity and detailed information of the original cough signal. This design choice minimizes the potential loss of critical acoustic features that may occur through downsampling or dimensional reduction at the input stage.

Following the final max-pooling layer, the resulting feature maps are flattened into a one-dimensional vector, which is subsequently fed into two fully connected (dense) layers. The first dense layer consists of 100 neurons to further abstract and integrate learned features, while the second dense layer comprises three neurons with a softmax activation function, enabling multiclass classification into COVID-19 positive, symptomatic non-COVID, and healthy categories.

For model optimization, the Adam optimizer was employed with an initial learning rate of 0.001. To enhance training stability and convergence, a learning rate decay schedule was implemented, gradually reducing the

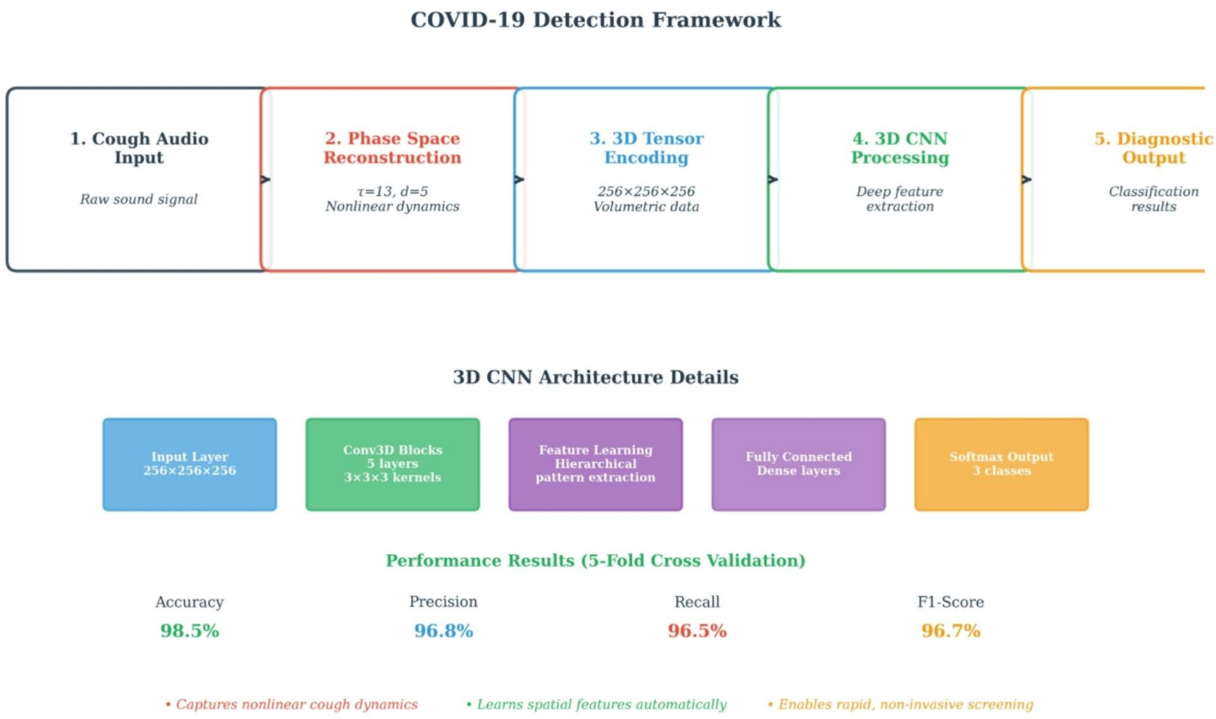

Fig. 3. COVID-19 detection framework workflow. The methodology processes cough audio through: (1) raw input acquisition, (2) PSR to capture nonlinear dynamics ($\tau = 13$, $d = 5$), (3) 3D tensor encoding for spatial representation, (4) 3D CNN architecture for hierarchical feature learning, and (5) diagnostic classification. The system achieves 98.5% accuracy through integrated nonlinear dynamic analysis and deep learning, enabling rapid, non-invasive COVID-19 screening.

learning rate to 0.0001 after 50 epochs. This progressive decay facilitates finer parameter tuning during later training stages, improving generalization performance and preventing premature convergence.

Where x represents the input to the activation unit. This non-linear function enables the model to capture complex relationships by activating only positive inputs, thereby promoting sparse representations and efficient gradient propagation during training.

All model development and experimentation were conducted on a high-performance workstation equipped with an AMD Ryzen 5 8645HS CPU, an NVIDIA GeForce RTX 4050 GPU featuring 8 GB of dedicated memory, and 8 GB of DDR5 RAM. This hardware configuration provided the necessary computational resources for efficient training and validation of the 3D DCNN model. The experimental environment was managed using Anaconda, with coding performed in Python 3.12 within the Spyder integrated development environment (IDE). The deep learning framework utilized was TensorFlow, complemented by the Keras API, enabling modular model construction and streamlined experimentation.

Results and discussion

The performance of the proposed model was rigorously assessed using a comprehensive suite of evaluation metrics, including classification accuracy, precision, recall (sensitivity), specificity, Area Under the Receiver Operating Characteristic Curve (AUC-ROC), and F1 score. These metrics collectively provide a balanced and thorough understanding of the model's discriminatory power, predictive reliability, and robustness in classifying cough audio into COVID-19 positive, symptomatic non-COVID, and healthy categories. Accuracy reflects the overall proportion of correctly classified instances and provides a general measure of effectiveness. Precision quantifies the proportion of true positive predictions among all positive predictions, indicating the model's ability to minimize false positives. Recall (Sensitivity) measures the model's capacity to identify true positive cases, which is critical in medical diagnosis to reduce false negatives. Specificity indicates the ability to correctly identify healthy or non-COVID cases, minimizing false alarms. AUC-ROC evaluates the model's performance across all classification thresholds, offering a comprehensive indicator of separability between classes. F1 Score serves as a harmonic mean of precision and recall, reflecting the balance between false positives and false negatives – particularly important in imbalanced datasets. In addition to these metrics, it is beneficial to report confusion matrices to visualize detailed classification outcomes and error types, and to compute balanced accuracy especially when dealing with imbalanced class distributions. Metrics like Matthews Correlation Coefficient (MCC) can provide further insight into model performance under such imbalance. Moreover, reporting confidence intervals or statistical significance tests (e.g., 95% confidence intervals for metrics, or p-values comparing with baseline models) enhances transparency and reliability of the evaluation. Lastly, presenting computational efficiency metrics, such as training time, inference latency, and model parameter counts, can be valuable to contextualize the feasibility of deploying the model in real-world scenarios, especially on resource-constrained devices as Eq. (4) to Eq. (8).

$$Acc = \frac{TP + TN}{TP + FP + FN + TN} \tag{4}$$

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Recall\,(sensitivity) = \frac{TP}{TP + FN} \tag{6}$$

$$specificity = \frac{TN}{TN + FP} \tag{7}$$

$$F1 = 2\frac{precision \times recall}{precision + recall} \tag{8}$$

True Positive (TP) refers to instances in which both the model's prediction and the actual condition are positive—for example, correctly identifying COVID-19 positive samples. Conversely, a False Positive (FP) occurs when the model predicts a positive outcome while the actual condition is negative, such as misclassifying healthy samples as COVID-19 positive. True Negative (TN) denotes cases where both the prediction and actual state are negative, for instance, correctly identifying healthy individuals. Finally, a False Negative (FN) indicates instances where the prediction is negative but the actual condition is positive, exemplified by failing to detect COVID-19 in infected subjects.

The evaluation of the proposed model was conducted using a rigorous 5-fold cross-validation protocol. Specifically, the dataset was partitioned into five mutually exclusive subsets of approximately equal size. In each fold, four subsets were utilized for model training, while the remaining subset served as the test set. This process was repeated five times, ensuring that each subset was used exactly once for testing. The final performance metrics were computed by averaging the results across all five folds, providing a robust estimate of the model's generalization capability.

Regarding model interpretability, the PSR representation itself offers a physical insight: the dispersed and complex attractor geometry observed in COVID-19 positive coughs (Fig. 2) likely corresponds to the irregular, chaotic dynamics of a respiratory system under stress from viral infection, potentially reflecting inflammation, mucus buildup, or altered vocal fold vibration. While the 3D DCNN effectively learns hierarchical features from this space, the specific features remain abstract. In future work, we plan to employ explainable AI (XAI) techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) adapted for 3D CNNs to identify which regions of the phase space tensor are most influential for the classification decision, thereby creating a direct link between the model's reasoning and the physiological characteristics of the cough.

To investigate the influence of the Reconstructed Phase Space (RPS) embedding dimension on the classification accuracy, a series of experiments were performed with dimensionalities ranging from 3 to 7. Notably, for the case of $d = 3$, the raw RPS representation was used without applying Principal Component Analysis (PCA), whereas for higher-dimensional cases ($d > 3$), PCA was employed to reduce the RPS data to three dimensions, facilitating compatibility with subsequent processing stages. Table 3 summarizes the classification performance of the proposed system across the different RPS dimensionalities, enabling a comparative analysis to identify the optimal embedding configuration consistent with theoretical expectations.

The results presented in Table 3 highlight the influence of the RPS embedding dimension on the classification performance of the proposed model, with particular attention to the role of dimensionality reduction via Principal Component Analysis (PCA). When using a raw 3-dimensional RPS representation ($d = 3$) without PCA, the model achieves a strong accuracy of 97.1% and an F1 score of 93.6%, demonstrating the efficacy of the direct low-dimensional representation. For higher embedding dimensions ($d = 4$ to 8), PCA is applied to reduce the dimensionality back to three, facilitating compatibility with the model input and mitigating the curse of dimensionality.

Optimal results are observed at an embedding dimension of five ($d = 5$) prior to PCA, where the model attains its highest accuracy of 98.5%, precision of 96.8%, recall of 96.5%, and F1 score of 96.7%. This indicates that the selected embedding captures sufficient dynamic information while PCA effectively preserves critical features

Dimension of RPS	Accuracy	Precision	Recall	Specificity	F1
d = 3	97.1	93.1	94.1	99.4	93.6
d = 4	96.6	91.8	93	99.3	92.4
d = 5	98.5	96.8	96.5	99.7	96.7
d = 6	96.7	89.8	93.3	99.1	91.5
d = 7	95.2	84.1	90.7	98.5	87.3
d = 8	90.7	70.8	82.2	96.9	76.1

Table 3. Results of proposed model using d-dimensional RPS representation, $d = 3, 4, 5, 6, 7$.

during dimensionality reduction. Beyond this optimal point, as the dimensionality increases further (e.g., d = 7), the model's performance declines—accuracy drops to 95.2% and the F1 score decreases to 87.3%—suggesting that excessive dimensional complexity or over compression through PCA may result in loss of informative signal characteristics essential for robust classification.

Figure 4 graphically depicts the trends in accuracy and F1 score across the range of tested RPS embedding dimensions, illustrating a clear performance peak at d = 5. Additionally, Table 4 presents confusion matrices corresponding to RPS representations of dimensions 3 through 8, providing detailed insights into classification errors and class-wise performance variability associated with each embedding dimension.

In comparison to prior studies, the proposed method demonstrates superior and more balanced performance across a comprehensive set of evaluation metrics based on provided details in Table 4. For instance, achieved an accuracy of 74.4% and an F1 score of 73%, although specificity was not reported, limiting assessment of false positive rates. Study reported improved accuracy of 79% and precision of 87%, yet its recall was notably low at 49%, indicating difficulties in correctly identifying positive COVID-19 cases and a tendency towards false negatives. Meanwhile, attained excellent results with 99.19% accuracy, 99% precision, 98% recall, and strong specificity of 95.05%, producing a robust F1 score of 95%. Similarly, achieved high accuracy of 97.59% and outstanding specificity of 98.73%, albeit with lower precision (88.34%) and F1 score (89.04%), suggesting some imbalance in classification performance. By contrast, our proposed method attains an accuracy of 98.5%, precision of 96.8%, recall of 96.5%, specificity of 99.7%, and an F1 score of 96.7%, consistently outperforming previous approaches—particularly with respect to recall and specificity. This balance indicates the model's strong capability to correctly identify COVID-19 positive cases while effectively minimizing false positives, thereby offering a reliable and robust solution for cough-based COVID-19 detection.

Ablation study

To evaluate the contribution of each component in our proposed framework, we conducted an ablation study. We compared the performance of our full model (PSR + 3D DCNN) against two baselines: (1) a 2D CNN model using standard Mel-Frequency Cepstral Coefficients (MFCCs) as input, and (2) our 3D DCNN architecture using a 3D tensor derived from a Short-Time Fourier Transform (STFT) spectrogram instead of PSR. As shown in Table 5, the model using MFCCs with a 2D CNN achieved respectable performance but was significantly outperformed by spectrogram-based 3D DCNN, highlighting the benefit of the 3D architecture for capturing spatiotemporal patterns. However, our full PSR-based model demonstrated a substantial improvement over the spectrogram-based approach, particularly in recall and F1-score. This confirms that the PSR is the key innovation, providing a more discriminative feature representation of the cough signal's nonlinear dynamics than traditional time-frequency analysis.

The results show that our PSR-based model (98.5% accuracy) substantially outperforms a 2D CNN using MFCCs (90.2% accuracy). This provides quantitative evidence that PSR captures more discriminative information for this task.

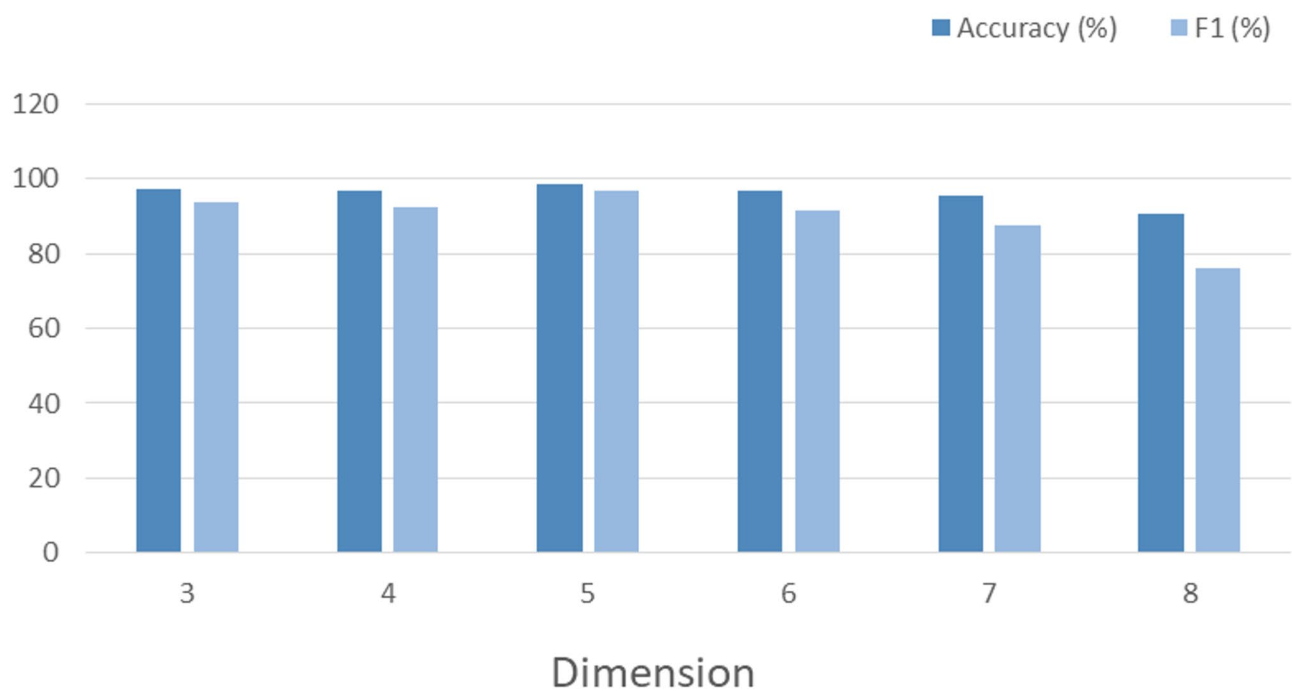

Fig. 4. Accuracy and F1 values over different dimensions.

d = 3	Covid 19	Healthy	Symptomatic
Covid 19	619	31	8
Healthy	39	6342	85
Symptomatic	7	75	1201
d = 4	Covid 19	Healthy	Symptomatic
Covid 19	612	34	12
Healthy	46	6317	103
Symptomatic	9	82	1192
d = 5	Covid 19	Healthy	Symptomatic
Covid 19	635	23	0
Healthy	21	6406	39
Symptomatic	0	43	1240
d = 6	Covid 19	Healthy	Symptomatic
Covid 19	614	38	6
Healthy	62	6307	97
Symptomatic	8	66	1209
d = 7	Covid 19	Healthy	Symptomatic
Covid 19	597	49	12
Healthy	93	6221	142
Symptomatic	20	89	1174
d = 8	Covid 19	Healthy	Symptomatic
Covid 19	541	94	23
Healthy	186	6017	263
Symptomatic	37	177	1069

Table 4. Confusion matrices are achieved by RPS representation.

Model configuration	Input representation	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)
Baseline 1: 2D CNN	MFCCs	90.2 ± 0.8	87.5 ± 1.2	85.9 ± 1.5	95.1 ± 0.6	86.7 ± 1.1
Baseline 2: 3D DCNN	STFT Spectrogram	94.7 ± 0.5	92.3 ± 0.9	91.8 ± 1.1	97.8 ± 0.4	92.0 ± 0.8
Proposed: 3D DCNN	PSR Tensor	98.5 ± 0.3	96.8 ± 0.6	96.5 ± 0.7	99.7 ± 0.2	96.7 ± 0.5

Table 5. Ablation study comparing the proposed method with baseline models. *Note: Results are reported as mean ± standard deviation across 5-fold cross-validation. MFCCs: Mel-Frequency Cepstral Coefficients; STFT: Short-Time Fourier Transform; PSR: PSR.*.

Conclusion

The study presents a novel framework for COVID-19 detection using cough audio signals by combining PSR and a custom 3D deep convolutional neural network. PSR transforms cough signals into multidimensional patterns, enabling effective feature extraction and classification into COVID-19 positive, symptomatic non-COVID, and healthy classes. Experiments on the COUGHVID dataset achieve 98.5% accuracy and 96.7% F1 score, outperforming existing methods in recall and specificity. Optimal results are obtained with RPS dimension five and PCA reduction. The framework offers a scalable, non-invasive diagnostic tool applicable to other respiratory diseases and future mobile health applications.

Despite the promising results, this study has several limitations. First, the model was trained and validated on the COUGHVID dataset, which, while large, is crowd-sourced and may contain label noise and variability in recording quality. Future validation on clinically validated, multi-center datasets is essential. Second, the computational complexity of the 3D DCNN and the PSR-to-tensor conversion is high, which could be a barrier for real-time deployment on mobile devices; future work will focus on model compression and optimization. Third, while PSR provides a rich feature space, the model's decision-making process is not fully transparent. Integrating explainable AI (XAI) techniques like Grad-CAM could help elucidate the link between phase space features and physiological characteristics.

Future research should explore the integration of additional respiratory sound modalities (e.g., breath and speech) into a multimodal framework and the deployment of lightweight, real-time models suitable for mobile health applications. Furthermore, the fusion of acoustic biomarkers with other clinical data could further enhance diagnostic accuracy and robustness.

The deployment of such a cough-based diagnostic system in real-world settings necessitates careful consideration of ethical and practical factors. Privacy and data security are paramount, as audio recordings are sensitive biometric data. Informed consent must be obtained, and data should be anonymized and stored securely.

Practically, the system should be presented as a screening and triage tool to assist healthcare professionals, not as a definitive diagnostic device, to manage expectations and prevent over-reliance. Regulatory approval from bodies like the FDA or CE would be required before clinical use. Finally, efforts must be made to ensure the model is fair and unbiased across different demographics, ages, and accents.

Data availability
The source code and preprocessed data supporting the findings of this study will be made publicly available on reasonable request from Dr. Abdolvahab Ehsani Rad, as Corresponding Author, and Email is vahabrad@iau.ir.

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Declarations

Competing interests
The authors declare no competing interests.

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